

Types

of neurons, neurotransmitters, and receptors

Computational Neuroscience

University of Bristol

M Rule

Learning outcomes:

- ▶ Recall and describe types of neurotransmitters and receptors
- ▶ !This deck is fact heavy.
 - I will put a **box around content** you should make flash cards for to prepare for the exam.
 - This may not be exhaustive, please cross-reference with emphasised content in coursenotes and tell me if I missed something.

Dale's principle

- ▶ **is:** Each neuron typically uses the same set of chemical neurotransmitters at all its axonal terminals.
- ▶ **is not:** All neurons are either excitatory or inhibitory

Dale's principle

Individual neurons typically use the same set of chemical neurotransmitters at all their axonal terminals.

Neurons can be categorised based on the chemical signals they send, the way they process inputs, and how they connect to other neurons.

Nevertheless, two (very broad) cell classes for this course

► **Excitatory** “E” cells:

- Release **glutamate** (among other things)
- Tend to excite their targets
- “Pyramidal” cells
- Most long-range projections are excitatory (though there are some long-range inhibitory connections)

► **Inhibitory** “I” cells:

- Release **GABA** (among other things)
- Tend to inhibit their targets (in mature animals)
- Various classes of local interneurons, some projection neurons

► ...many others

Receptors are

Ionotropic

- ▶ Neurotransmitter binding changes an ionic conductance

Metabotropic

- ▶ Neurotransmitter binding triggers a chemical signal
- ▶ (In some cases this chemical signal can indirectly change ionic conductances)

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Not to be confused with *channels*, which may be one or more of

Ligand-gated

- ▶ This is the same group as ionotropic receptors

Voltage-gated

- ▶ Conductance responds to the membrane voltage

...and several other gating mechanisms

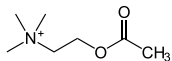
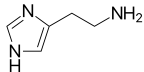
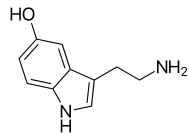
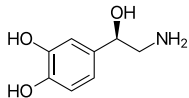
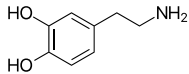
Glutamate receptors:

- ▶ **AMPA:** Ionotropic, Na^+ and K^+
 - -Amino-3-hydroxy-5-methyl-4-isoxazolepropionic
 - (limited medicinal drugs, some compounds used in experimental neuroscience)
- ▶ **NMDA:** Ionotropic, Ca^{2+}
 - N-methyl-D-aspartic acid
 - Ketamine, Dextromethorphan, PCP, N_2O , Memantine
- ▶ **mGluRs:** Metabotropic
 - (limited medicinal drugs, some compounds used in experimental neuroscience)

GABA receptors:

- ▶ **GABA_A:** Ionotropic, Cl^-
 - Benzodiazepines act here
- ▶ **GABA_B:** Metabotropic
 - GHB: replace the NH_2 in GABA with with OH

Neuromodulators



Dopamine

- ▶ (all metabotropic)

Norepinephrine

- ▶ (all metabotropic)

Serotonin

- ▶ 5HT₃ ionotropic (Na^+ and K^+ Ca^{2+})
- ▶ (all others metabotropic)

Histamine

- ▶ (all metabotropic)

Acetylcholine

- ▶ Nicotinic nAChRs are ionotropic Na^+ and K^+ (some Ca^{2+})
- ▶ Muscarinic (mAChR) are metabotropic

ligand/voltage-gated, EPSP/IPSP

end

... Quantum Chemistry, Molecular dynamics

Molecules

Gillespie, Master Equation

Concentrations

Mass-Action Kinetics

Conductance Models

Hodgkin-Huxley

Spiking Models

Leaky Integrate and Fire

Rate Neurons

Neural Mass/Field Models

Poisson Neurons

Generalized Linear Models

Binary Neurons

McCulloch-Pitts, Hopfield, Perceptron

Cognitive Neuroscience, Psychology ...

Physiological,
Quantitative

Biological
Realism,
Data needed
to identify
parameters



Computational
Efficiency,
Mathematical
Tractability



Phenomenological,
Qualitative